Matrices & Matrix Algebra

Section 1: Matrix Multiplication Properties

1 If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2

b. 4×4

c. 2×3

d. Multiplication is undefined

2 Is matrix multiplication commutative in general ($AB = BA$)?

a. No, $AB \neq BA$ in general even when both products are defined

b. Yes, multiplication of numbers and matrices is always commutative

c. Only if the entries are integers

d. Yes, provided both matrices are square

3 What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?

a. $B^T A^T$

b. $A^T B^T$

c. AB^T

d. BA^T

Section 2: Matrix Inverses and Elementary Matrices

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4 Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

5 What is an elementary matrix?

a. A matrix obtained by performing a single elementary row operation on an identity matrix

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

6 If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

Section 3: Symmetric, Orthogonal, and Special Matrices

7 A square matrix A is symmetric if and only if:

a. $A^T = A$

b. $A^T = -A$

c. $A^T = A^{-1}$

d. $\det(A) = 1$

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8 What defines an orthogonal matrix Q?

a. $Q^TQ = I$, meaning $Q^{-1} = Q^T$

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

9 If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1

b. 0

c. 1 only

d. Any real number

Section 4: Linear Transformations and Standard Matrices

10 A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = \|\mathbf{u}\|$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

11 What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

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12 What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$

b. $[[-1, 0], [0, 1]]$

c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

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